Statistical Mechanics (MP461) Assignment 2

1. Molar masses

Calculate the mass of a mole of dry air, which is a mixture of $N_2(78\%)$, $O_2(21\%)$ and argon (1%).

Solution

- 78% N_2 : 2×14.0067 g/mol
- 21% O_2 : 2×15.999 g/mol
- 1% Ar : 2×39.948 g/mol

Total is then

$$2 \times 0.78 \times 14.0067 + 2 \times 0.21 \times 15.995 + 0.01 \times 39.948 = 28.9695\text{g/mol}$$

2. Heat capacity of water

The heat capacity of 0.125kg of water is measured to be 523J/K at room temperature.
(i) Use this to calculate the heat capacity of water per unit mass (e.g. the *specific* heat capacity).
(ii) What is the molar mass of water? Use this to calculate the *molar* heat capacity of water.

Solution

(i) Specific heat capacity is $c = C/m = \frac{523\text{J/K}}{0.125\text{kg}} = 4184\frac{\text{J}}{\text{K}\times\text{kg}} = 4.184\frac{\text{J}}{\text{K}\times\text{g}}$

(ii) To get the molar heat capacity we multiply the specific heat capacity by the mass per mole of water which in this case is about 18g/mol. This gives $c_{mol} = 4.184\frac{\text{J}}{\text{K}\times\text{g}} \times 18\frac{\text{g}}{\text{mol}} = 75.32\frac{\text{J}}{\text{K}\times\text{mol}}$

3. Heat Exchange

Two bodies, with heat capacities C_1 and C_2 (assumed independent of temperature) and initial temperatures T_1 and T_2 respectively, are placed in thermal contact. (i) Show that their final temperature T_f is given by $T_f = (C_1T_1 + C_2T_2)/(C_1 + C_2)$. (ii) If C_1 is much larger than C_2, show that $T_f \approx T_1 + C_2(T_2 - T_1)/C_1$.

Solution

(i) Conservation of Energy gives $E_i = E_f$.. Since the only relevant energy is heat this means that

$$(C_1 + C_2)T_f = C_1T_1 + C_2T_2$$
$$T_f = \frac{C_1T_1 + C_2T_2}{C_1 + C_2}$$

(ii) Rewriting and doing a Taylor expansion around 0 we get

$$T_f = T_1 \frac{1 + \frac{C_2}{C_1}\frac{T_2}{T_1}}{1 + \frac{C_2}{C_1}}$$

This form can be expanded according to

$$\frac{1 + ax}{1 + x} = 1 + (a - 1)x + (1 - a)x^2 + O(x^3)$$

to get

$$\begin{aligned} T_f &= T_1 \left[1 + (\frac{T_1}{T_2} - 1)\frac{C_2}{C_1} + O(x^2)\right] \\ &\approx T_1 + (T_2 - T_1)\frac{C_1}{C_2} \end{aligned}$$

4. Exponential atmosphere

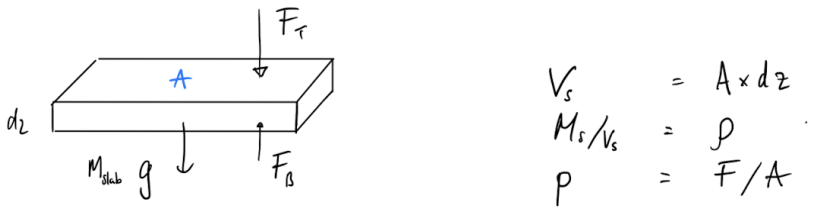
(a) Consider a horizontal slab of air whose thickness is dz. If this slab is at rest, the pressure holding it up below must balance both the pressure from above and the weight of the slab. Use this fact to find an expression for dp/dz, the variation of pressure with altitude in terms of the density of the air.

(b) Use the ideal gas law to write the density of air in terms of pressure, temperature, and the average mass m of the air - see Problem 1 above. Show then that the pressure obeys the barometric equation

$$\frac{dp}{dz} = -\frac{mg}{k_BT}p \tag{1}$$

(c) Assuming that the temperature of the atmosphere is independent of height (not a great assumption but not terrible either), solve the barometric equation to obtain the pressure as a function of height.

(d) Find a similar expression for the number density n as a function of height.



Solution (a) The basic setup is shown in the figure. We need to balance all the forces so we have

$$\begin{aligned} F_B - F_T - m_s g &= 0 \\ \frac{F_B}{A} - \frac{F_T}{A} - \frac{m_s g}{A} &= 0 \\ p(z) - p(z+dz) - \frac{m_s g}{A} &= 0 \end{aligned}$$

where A is the area of the slab and $p = F/A$ is the pressure. We rewrite this by dividing by dz to get

$$\frac{p(z+dz) - p(z)}{dz} = -\frac{m_s g}{A dz} = -\frac{m_s g}{V_s} = -\rho g$$

In the limit that $dz \to 0$ we have then that

$$\frac{dp}{dz} = -\rho g \tag{2}$$

(b)We use the ideal gas equation to get $\frac{p}{k_B T} = \frac{N}{V}$ and then observe that

$$\rho = \frac{m_s}{V_s} = \frac{m N_s}{V_s} = \frac{mp}{k_B T}$$

where m is the average molecular mass. This allows us to rewrite this as

$$\frac{dp}{dz} = -\frac{mp}{k_B T} g. \tag{3}$$

(c) To solve this set $c = \frac{mp}{k_B T}$ and rewrite to get

$$\frac{dp}{p} = -c dz$$

we then integrate from height 0 to h to get

$$\int_{p(0)}^{p(h)} \frac{dp}{p} = -c \int_0^h dz$$

Integrating both sides and evaluating limits we get

$$\ln p(h)/p(0) = -ch$$

Taking exponential of each side and rearraging gives

$$p(h) = p(0)e^{-mgh/k_BT}$$

(d) To relate pressure to number density we just use that $n = p/k_BT$ and so at constant temperature we can just substitute n for p

$$n(h) = n(0)e^{-mgh/k_BT}$$

5. Better calculations through Boltzmann

Use your understanding of Boltzmann factors to come up with the same expression [as Q4 (d)] for $n(z)$.

Solution We can assume that the probability of finding a particle at a height h is proprtional to its number density at that point. If plug $E = mgh$ into the Boltzmann expression $P(h) \propto e^{-E/k_BT}$ we get

$$n(h) \propto P(h) \propto e^{-mgh/k_BT}$$

as in Q4 d above.

6. A three-level atom

An atom can be in one of three states, with energies 0 eV, 0.2 eV and 0.4 eV. The atom is at thermal equilibrium with a reservoir at 2500 K. Calculate (a) the probability for this atom to be in each of the three states, and (b) the average energy of the atom.

Solution

1. With $T = 2500\,\mathrm{K}$, we have $k_BT = 5.524 \cdot 10^{-20}\mathrm{J} = 0.2154\,\mathrm{eV}$. The Boltzmann weights for the three states are

$$\begin{aligned} e^{-E_0/k_BT} &= e^{-0.0/0.2154} = 1.0\,, \\ e^{-E_1/k_BT} &= e^{-0.20/0.2154} = 0.3952\,, \\ e^{-E_2/k_BT} &= e^{-0.40/0.2154} = 0.1562\,. \end{aligned}$$

   The sum of the Boltzmann weights is $Z = 1.0 + 0.3952 + 0.1562 = 1.5514$, and the probabilities of the three states are hence

$$\begin{aligned} P(E_0) &= \frac{1.0}{1.5514} \underline{= 0.64} \\ P(E_1) &= \frac{0.3952}{1.5514} \underline{= 0.25} \\ P(E_2) &= \frac{0.1562}{1.5514} \underline{= 0.10} \end{aligned}$$

2. The average energy (or the expectation value of the energy) is given by

$$\langle E \rangle = \sum_n E_n P(E_n) = (0.64 \cdot 0 + 0.25 \cdot 0.2 + 0.10 \cdot 0.04)\,\text{eV} \underline{= 0.09\,\text{eV}\,.}$$

7. Macrostates and microstates

A system consists of a set of uniformly spaced energy levels of separation $1 \times 10^{-20} J$, starting from the first level which has an energy of $0J$. Eight distinguishable particles are placed within this system with a total energy of $6 \times 10^{-20} J$.

(a) How many possible configurations are there satisfying this condition?
(b) Which of these is the most probable?

Solution With the notation # of particles $\times$ energy level the number of ways of adding to a total energy of $6 \times 10^{-20} J$ is

1. $1 \times 6 + 7 \times 0$
2. $1 \times 5 + 1 \times 1 + 6 \times 0$
3. $1 \times 4 + 1 \times 2 + 6 \times 0$
4. $1 \times 4 + 2 \times 1 + 5 \times 0$
5. $2 \times 3 + 6 \times 0$
6. $1 \times 3 + 1 \times 2 + 1 \times 1 + 5 \times 0$
7. $1 \times 3 + 3 \times 1 + 4 \times 0$
8. $3 \times 2 + 5 \times 0$
9. $2 \times 2 + 2 \times 1 + 4 \times 0$
10. $1 \times 2 + 4 \times 1 + 3 \times 0$
11. $6 \times 1 + 2 \times 0$

The probability of a particular microstate occurring has to do with how much energy it has. However, these are macrostates with many associated microstates. We want to count how many we use the general formula

$$\Omega = \frac{N!}{\prod_i n_i!} \tag{4}$$

1. $\frac{8!}{(\prod(0!)1!7!} = 8$

2. $\frac{8!}{\prod(0!)1!1!6!} = 56$
3. $\frac{8!}{\prod(0!)1!1!6!} = 56$
4. $\frac{8!}{\prod(0!)1!2!5!} = 168$
5. $\frac{8!}{\prod(0!)2!6!} = 28$
6. $\frac{8!}{\prod(0!)1!1!1!5!} = 336$
7. $\frac{8!}{\prod(0!)1!3!4!} = 280$
8. $\frac{8!}{\prod(0!)3!5!} = 56$
9. $\frac{8!}{\prod(0!)2!2!4!} = 420$
10. $\frac{8!}{\prod(0!)1!4!3!} = 280$
11. $\frac{8!}{\prod(0!)6!2!} = 28$

The largest probability = The largest multiplicity =420.

8. A question of speed

Show that for a Maxwell-Boltzmann velocity distribution $v_{max} < \langle v \rangle < v_{rms}$.

Solutions

- (v_{max}):
  Since v_{max} is the maximum of the Maxwell Boltzmann probability distribution

$$P(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv/2k_B T}$$

  we take the derivative and set to zero. We get

$$\frac{dp}{dv} = 4\pi \left(\frac{m}{2\pi k_b T}\right)^{3/2} \left[2ve^{-mv^2/2k_B T} - \frac{m}{2k_B T} e^{-mv^2/2k_B T} 2v^3\right] = 0$$

  Clearly $v = 0$ works here but that is v_{min}. For v_{max} we have that $1 - \frac{mv^2}{2k_b T} = 0$ and so

$$v_{max} = \sqrt{\frac{2k_B T}{m}}$$

- ($\langle v \rangle$):

  The mean velocity $\langle v \rangle$ is given by the integral:

$$\langle v \rangle = \int_0^\infty v \cdot P(v)\, dv = \int_0^\infty v \cdot 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}}\, dv$$

  Simplifying the integral:

$$\langle v \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty v^3 e^{-\frac{mv^2}{2kT}}\, dv$$

  We then make the substitution $u = -\frac{mv^2}{2kT}$ and $du = -\frac{mv}{kT}\, dv$:

$$\langle v \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty \left(-\frac{2kT}{m}u\right) e^u \left(-\frac{kT}{m}\right)\, du$$

  Simplifying further:

$$\langle v \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(\frac{2kT}{m}\right)\left(\frac{kT}{m}\right) \int_0^\infty u e^u\, du$$

  Since $\int_0^\infty ue^u\, du$ is a standard result equal to 1 we have then that

$$\langle v \rangle = \sqrt{\frac{8kT}{\pi m}}$$

- v_{rms}:

  For $v_{rms} = \sqrt{\langle v^2 \rangle}$ we can compute the integral (and others like it) by noting

$$\int_{-\infty}^\infty e^{-ax^2} dx = 2\int_0^\infty e^{x^2} \sqrt{\frac{\pi}{a}} \tag{5}$$

$$\frac{d}{da}\int_0^\infty e^{-ax^2} dx = -\int_0^\infty x^2 e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}\frac{d}{da}a^{\frac{-1}{2}} = -\frac{\sqrt{\pi}}{4}a^{\frac{-3}{2}} \tag{6}$$

$$\frac{d^2}{da^2}\int_0^\infty e^{-ax^2} dx = \int_0^\infty x^4 e^{ax^2} dx = -\frac{3\sqrt{\pi}}{8}a^{\frac{-5}{2}} \tag{7}$$

  With $a = \frac{m}{2k_BT}$ we find

$$< v^2 >= \frac{4\pi}{\pi^{\frac{3}{2}}} \cdot \frac{3\sqrt{\pi}}{8}\left(\frac{m}{2k_BT}\right)^{\frac{-5}{2}} = \frac{3}{2}\cdot\frac{2k_BT}{m} = \frac{3k_BT}{m} \tag{8}$$

  So $v_{rms} = \sqrt{\frac{3k_BT}{m}}$

9. Equipartition Theorem

(a) If the energy of a classical system is a sum of n quadratic modes, i.e.

$$E = \sum_{i=1}^{n} \alpha_i x_i^2, \tag{9}$$

show that if the system is in contact with a heat reservoir at temperature T, the mean energy of the system is given by

$$\langle E \rangle = n \times \frac{1}{2} k_B T \tag{10}$$

Solution

$$\text{if suppose} \quad E = \sum_{i=1}^{n} \alpha_i x_i^2 \quad \Rightarrow \quad \langle E \rangle = \sum_{i=1}^{n} \alpha_i \langle x_i^2 \rangle$$

$$\langle E \rangle = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} dx_1 dx_2 \ldots dx_n \quad \sum_{i=1}^{n} \alpha_i x_i^2 \quad P(x_1, x_2 \ldots x_n)$$

$$\langle E \rangle = \frac{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} dx_1 \ldots dx_n \quad \sum_{i=1}^{n} \alpha_i x_i^2 \quad e^{-\sum \alpha_i x_i^2 / k_B T}}{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} dx_1 \ldots dx_n \quad e^{-\sum \alpha_i x_i^2 / k_B T}}$$

$$= \frac{\sum_{i=1}^{n} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} dx_1 \ldots dx_n \quad \alpha_i x_i^2 \quad e^{-\sum \alpha_i x_i^2 / k_B T}}{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} dx_1 \ldots dx_n \quad e^{-\sum \alpha_i x_i^2 / k_B T}}$$

examine $i = j$

$$\frac{\int_{-\infty}^{\infty} dx_1 e^{-\alpha_1 x_1^2 / k_B T} \ldots \int_{-\infty}^{\infty} dx_j \, \alpha_j x_j^2 \, e^{-\alpha_j x_j^2 / k_B T} \ldots \int_{-\infty}^{\infty} dx_n e^{-\alpha_n x_n^2 / k_B T}}{\int_{-\infty}^{\infty} dx_1 e^{-\alpha_1 x_1^2 / k_B T} \ldots \int_{-\infty}^{\infty} dx_j \, e^{-\alpha_j x_j^2 / k_B T} \ldots \int_{-\infty}^{\infty} dx_n e^{-\alpha_n x_n^2 / k_B T}}$$

$$= \sum_{i=1}^{n} \frac{\int dx_j \, \alpha_j x_j^2 \, e^{-\alpha_j x_j^2 / k_B T}}{\int dx_j \, e^{-\alpha_j x_j^2 / k_B T}} = \frac{n}{2} k_B T$$

Equipartition theorem
If the energy of a classical system is the sum of n quadratic modes, and that system is in contact with a heat reservoir at temperature T, the mean energy of the system is given by $n \times \frac{1}{2} k_B T$.

***10.Heat capacity and equipartition**

Consider a crystalline solid consisting of N particles of mass m. Use the equipartition theorem to derive that, at sufficiently high temperatures, the molar specific heat of such a solid equals 3R, where R is the gas constant.

Solution We model a solid as $3 \times N$ Harmonic oscillators. Each oscillator has energy $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ and so we have $6N$ quadratic modes. The equipartition theorem says then that

$$U = N\langle E\rangle = 6N \times \frac{1}{2}k_BT = 3Nk_BT$$